

Yunhua Zhao

Department of Computer Science
The Graduate Center, CUNY
New York, New York, 10016
United States of America

Contact

Phone: +1 (646) 645-4632
Email: yzhao5@gradcenter.cuny.edu
 [Google Scholar](#)

Education

City University of New York, Graduate Center

Ph.D. in Computer Science

2019–2026

Advisor: Professor Hui Chen

GPA: 3.76/4.0

Thesis: "Application of Machine Learning for Software Quality Assurance"

North China Electric Power University

M.S. in Signal and Information Processing

2013–2016

Advisor: Professor Lianqing Zhao

Thesis: "Plateless Digital Printing"

North China Electric Power University

Bachelor in Electric and Information Engineering

2009–2013

Lecturer and Teaching Assistant Experience @ CUNY

Lecturer of City College, City University of New York

Artificial Intelligence

Visualization

Senior Design

A Data Science and Statistical Approach to Programming

Lecturer of Brooklyn College, City University of New York

Discrete Mathematics

Design and Implementation of Large Scale Web Application

Introduction to Modern Programming Techniques (Java)

Introduction to Programming Using Python

Lecturer of John Jay College, City University of New York

The Nature of Computers and Computing

TA of Graduate Center, City University of New York

Machine Learning

Data Mining

Publications

1. [Zhao, Yunhua](#) et al. (2023). "A Systematic Survey of Just-in-Time Software Defect Prediction". In *ACM Computing Surveys* 55.10, pp. 1–35. (Citation: 95+)
2. [Zhao, Yunhua](#) and Hui Chen (2023). "Deep Incremental Learning of Imbalanced Data for Just-In-Time Software Defect Prediction".
3. Hui Chen, [Yunhua Zhao](#), Kostadin Damevski (2025). "Improving Data Curation of Software Vulnerability Patches through Uncertainty Quantification". *IEEE Transactions on Software Engineering*.

Ph.D. Research: Application of Machine Learning for Software Quality Assurance

Research Focus: Specializing in the application of Machine Learning to Software Engineering, with an emphasis on developing AI-driven tools that enhance software quality assurance and support developers across the software lifecycle. Current work leverages Large Language Models (LLMs), Reinforcement Learning (RL), and Deep Neural Networks (DNNs) to address complex challenges in software development,

maintenance, and automated defect prediction.

- Conducting research on interactive models for software quality assurance using *reinforcement learning (RL)* to improve software quality.
- Developing applications of *Large Language Models (LLMs)*, including *encoder, decoder* and *transformer* techniques, to enhance code understanding and quality assessment.
- Built *LSTM* based *time-series* models for just-in-time software defect prediction (JIT-SDP).

Research Internships and Work Experience

Data Analysis Internship @ EMW3

Summary: Developed automated tools to extract and process data from various sources using the Google Cloud Platform (GCP), replacing manual operations.

04/2023–
06/2023

Student Researcher of Cyber Security @ RFCUNY

Advisor: Professor Adam Scott Wandt

Topic: "Designed and implemented protocols to gather data from IoT devices, networks"

04/2022–
07/2023

Student C++ Developer of HIDS @ RFCUNY

Summary: Developed C++ software for a Stochastic Computing (SC)-based Host Intrusion Detection System (HIDS) that extracts features from network packets and feeds them into a neural network for analysis.

09/2020–
01/2021

Software Engineer @ Ericsson (Full-time)

Summary: Developed and maintained core system components of switch and router operating systems to ensure stable process communication and reliable distributed database operations.

09/2017–
07/2019

Software Engineer @ New H3C Technologies (Full-time)

Summary: Conducted research and development in Software-Defined Networking (SDN), designing and implementing network communication protocols—including OpenFlow and custom flow control algorithms—for routers and switches at the socket communication level.

07/2016–
09/2017

Scholarship and Synergistic Activities

- **Reviewer** of Automated Software Engineering
- CUNY Doctoral Student Research Grant Fellowships, 05/2022
- CUNY Spring 2022 Open Knowledge Fellowship, 03/2022
- CUNY Provost's Pre-Dissertation Science Research Fellowships, 05/2021
- Served for conference ICSE 2022 as a Student Volunteer
- Served Executive Committee for the CUNY Graduate Center Computer Science Department
- **Presented** research paper and served as a panelist at the CUNY Celebrates Women in Computing Virtual Conference, May 7th, 2021